OAKLAND COMMUNITY COLLEGE

ROBOTICS/AUTOMATED SYSTEMS TECHNOLOGY

Course Number and Title: ROB 2140 Advanced Programmable Controller Appl.
Section Number:
Meeting Days and Times: Day, 5:00 PM to 9:50 PM
Classroom Location: T4
Term and Year: Fall 2025 AH1501

Instructor Name: John P. Sefcovic
Instructor Contact:
E-mail: jpsefcov@oaklandcc.edu

Textbook: Handouts will be available.
Other needed supplies: Optional: USB Flash Drive

Course Description: The course will be structured to provide the student with an understanding of the relationship between "real time" control systems and industrial devices and machines. The advanced instruction set of programmable controllers will be studied relevant to concepts and structures of automated control systems. Various applications will be defined in which the student will develop the written programs for each hardware and software specification of the process problems, including field devices, data networks, and Human Machine Interfaces (HMI). The use of the Robotics Lab equipment will give the student practical programming and troubleshooting skills used in the maintenance of automated systems

Common Course Outcomes and Course General Education Outcomes:

Common Course Outcome	GE Outcome
The student will be able to apply in Studio 5000 and SIEMENS TIA Step 7 Ladder Logic Data instructions in application programs.	Critical Thinking
The student will be able to apply in Studio 5000 and SIEMENS TIA Step 7 Function Block instructions in application programs.	Critical Thinking
The student will be able to apply in Studio 5000 and SIEMENS TIA Step 7 Structured Text instructions in application programs.	Critical Thinking
The student will define the parameters for Studio 5000 Add On Instructions and apply the instruction in application programs.	Critical Thinking
The student will be able to demonstrate the User Defined Data in Studio 5000 and SIEMENS TIA Step7.	Critical Thinking
The student will be able to demonstrate the programming of the SIEMENS TIA Step7 Reusable Data Blocks in an application.	Critical Thinking
The student will define and apply in Studio 5000 and SIEMENS TIA Step 7 analog concepts in application programs.	Quantitative Literacy Critical Thinking

The student will be able to develop and implement the same Human Machine Interfaces (HMI) applications with FactoryTalk View ME HMI and SIEMENS WinCC .	Critical Thinking

Course Goals:

- Explain the program and data structures used in Studio 5000 and SIEMENS TIA Step 7 programmable logic controller applications.
- Create programs in Studio 5000 Ladder Logic using the advanced instructions in an application.
- Create programs in SIEMENS TIA Step 7 Ladder Logic using the advanced instructions in an application.
- Explain and demonstrate the use of analog data in Studio 5000 and SIEMENS TIA Step7.
- Demonstrate the programming of the Studio 5000 and SIEMENS TIA Step 7 Function Block instructions in an application.
- Demonstrate the programming of the Studio 5000 and SIEMENS TIA Step 7 Structured Text instructions in an application.
- Explain and demonstrate the User Defined Data in Studio 5000 and SIEMENS TIA Step7.
- Demonstrate the programming of the Studio 5000 Add On Instructions in an application.
- Demonstrate the programming of the SIEMENS TIA Step7 Reusable Data Blocks in an application.
- Develop and implement Human Machine Interfaces (HMI) with FactoryTalk View ME HMI and SIEMENS WinCC.

Grading Standards: See Basis Of Course Grade, Grade Breakdown, Labs, and Midterm and Final below. Student-initiated withdrawals will not be permitted beyond the twelfth week of instruction.

Incomplete: This faculty initiated mark will be used sparingly and only when an emergency prevents a student from completing course work during the regular College session. The student is responsible for completing a written agreement with the instructor detailing the requirements to be met for the completion of the "I" before it is assigned. This grade may be granted only when the student has completed at least 70% of the classwork with a passing grade of “C” or higher. The student is not to register for a course in which he or she has a current mark of "I". Without prior faculty-initiated action to change the "I", this mark will become a "WS" one year subsequent to its original issue. Marks of "I" do not satisfy prerequisites and are not transferable. This mark is not used in the calculation of GPA, but it may affect the eligibility for financial aid.

N Mark (Grade) Non-Attendant: This mark is awarded to students who, though registered, never attended class and did not officially drop. Oakland Community College is adopting the Federal policy language, which is:

- For synchronous classes (whether on-site or online), "attendance" is determined by attendance in a class, lecture, field activity, recitation, or laboratory activity.
- For online asynchronous classes, "attendance" is determined by submission of gradable work.

Marks of "N" do not satisfy prerequisites and are non-transferable. This mark is not used in the calculation of GPA, but it may affect eligibility for financial aid. Instructors confirm non-attendance immediately after the Census Date (found on the Section Deadline Dates website for each section, https://www.oaklandcc.edu/section-dates) and then report non-attendance to the institution. Students who have received an "N" mark will not be permitted to enter the class, nor will the student be able to drop the class. After the "N" mark has been received, a change of grade will not be accepted for the student.

Attendance Policy: Attendance is mandatory for lecture and for the content delivered in the lecture. NO LECTURE WILL BE GIVEN FOR MISSED CONTENT. Attendance will be taken at the start of class, if the student is late it is their responsibility to notify the instructor.

Course Content:

- Studio 5000 Programming
    - Programming Structures
        - Tasks
        - Programs
        - Routines
    - Module Data Structures
    - Move and Logical Instructions
    - Arithmetic Instructions
    - Comparison Instructions
    - Program Control Instructions
    - Sequencer Instructions
    - Bit Shift, FIFO, and LIFO Instructions
    - File Manipulation Instructions
    - Analog Data and Scaling
    - File Instructions
    - Add-on Instructions
    - User Defined Data Type
    - Function Block Programming
    - Structured Text Programming
- Siemens TIA Portal Step 7
    - Programming Structures
        - Organization Block

        - Functions
        - Function Blocks
    - Move and Logical Instructions
    - Arithmetic Instructions
    - Comparison Instructions
    - File Manipulation Instructions
    - Analog Data and Scaling
    - Function Block Programming
    - Structured Text Programming
- RS FactoryTalk View ME HMI Programming
    - Creating Applications
    - Open Existing Applications
        - Communications Setup
        - Project Settings
        - Modify Graphic Displays
        - Objects
            - Drawing
            - Pushbutton
            - Indicators
            - Numeric and String
            - Gauge and Graph
            - Navigation
            - Object Manipulation
            - Properties
            - Tags
            - Alarms
                - Editor
                - Trigger
                - Display
            - Startup Setting
            - Runtime Files

FERPA Statement: Per the Family Educational Rights and Privacy Act (FERPA), college personnel are not allowed to release a student's personal information to anyone, including other students. For a full description of Oakland Community College's FERPA policy, please refer to: https://www.oaklandcc.edu/student-services/college-policies/default.aspx

Americans with Disabilities Act (ADA): Students requiring special assistance (including those affected by the Americans with Disabilities Act) should contact the ACCESS office, which will inform the instructor of any special accommodations pertaining to their learning.

Technology Appropriate Use Regulations: To access the Oakland Community College's policy on technology, please visit: https://www.oaklandcc.edu/student-services/college-policies/default.aspx

Public Safety: Oakland Community College has uniformed police officers at all academic locations. Please visit https://www.oaklandcc.edu/student-services/college-policies/default.aspx

College Closing and Cancelled Class Session:
Announcement of College Closings
To ensure the most effective and quickest way of finding out about a college or campus closing, be sure to enroll in OCC Emergency Alert at http://www.oaklandcc.edu/ENS. In the event of college or campus closings, an announcement will be made through major radio and television stations. ***Note this does not include a cancelled session.***

Announcement of Cancelled Class Session
If a class session is to be cancelled, due to illness or other circumstances, an email will be sent to your college registered account. If you do not monitor your college e-mail account, you may want to forward or add to an e-mail program you frequently monitor.

Changes to Scheduled Content due to Closing or Cancellation.
Changes, if any, will be announced in the session following the closing or cancellation. DO NOT CALL OR E-MAIL REQUESTING INFORMATION ABOUT CHANGES.

Grading Standards: See Basis Of Course Grade, Grade Breakdown, Labs, and Midterm and Final below.

Basis of Course Grade:

A	100-95	C+	77 - 79
A-	92 - 94	C	70 - 76
B+	89 - 91	C-	66 - 69
B	83 - 88	D+	60 - 65
B-	80 - 82	D	51 - 59
		F	- 50

WARNING: If you do not withdraw or make arrangements with the instructor to withdraw you from the course, *YOUR COURSE GRADE WILL BE DETERMINED FROM THE GRADES EARNED.* Student-initiated withdrawals will not be permitted beyond the twelfth week of instruction.

Grade Breakdown:

Labs	60%
Final Exam	40%

Attendance Policy: Attendance is mandatory for lecture and for the content delivered in the lecture. NO LECTURE WILL BE GIVEN FOR MISSED CONTENT.

Class Attendance / Tardiness: Students are expected to attend scheduled classes and to be on time. Failure to do this will be reflected in your final grade and in job recommendations. Attendance will be taken at the start of class, if the student is late, it is their responsibility to notify the instructor

Online Course Site:

The online course site will be used for the following:

Accessing the PowerPoint sides used during lecture for the labs.

The online course site ***will not be used*** for the following:

Quizzes
Discussions
E-mail
Dropbox
Checklists

Labs Assignments:

Labs will be assigned on material that has been discussed in the lecture.
See LAB SIGN-OFF SHEET for assignments.

Labs are assigned points towards the course grade. See LAB SIGN-OFF SHEET for points assigned.

ALL LABS ARE DUE AT THE END OF THE SESSION 14, December 2.

Final:

NO MAKE UP WILL BE GIVEN FOR THE FINAL UNLESS YOU HAVE CONTACTED THE INSTRUCTOR FOR ALTERNATIVE ARRAIGNMENTS BEFORE SESSION 14.

Final is scheduled for Session 15, December 9.

Lab problems for the Final are performance exams where programs will be assigned to be completed within the allotted time limit.

The Final will begin at the specified start time for the time limit of the exam.

The final is a performance exam with problems similar to the Studio 5000, FactoryTalk HMI, and Siemens labs. Programs for the performance Final are to be completed within the allotted time limit. The average of points earned will be applied to thirty percent (40%) of your Course Total Grade. Grade per program will be as follows:

- Complete and correct program will receive 100 points.
- A program with logic errors will receive between 95 or less points depending on the number of logic errors.

Testing policy: Exams can be made up if arrangements have been made by the student prior to the examination date.

Technical Issues:
If you cannot find the files location, the file was not loaded on the computer, or the file fails to open, notify the instructor immediately. If you fail to notify the instructor of any technical issue, it does not constitute a reason for not completing the lab assignment.

External Storage Devices:
External storage devices (USB drives) are permitted in the lab.

Back up your assignments at the end of each lab session. This includes files on the computer's hard drive and program files from the robot.

It is highly recommended you create a duplicate of your work as a backup on your home computer.

External storage devices that are lost or damaged does not constitute an excuse for missing or uncomplete assignments.

Class Attendance / Tardiness: Students are expected to attend scheduled classes and to be on time. Failure to do this will be reflected in your final grade and in job recommendations.

Lab: Students must report any problems to OCC robotics staff.

Behavior: Disruptive or abusive behavior is not allowed in lecture or lab and subject to college discipline policy.

Students may be dismissed or failed in this class for violating safety regulations, failure to complete lab assignments, or failing their exams.

This syllabus is subject to revision as deemed necessary by the instructor. Appropriate notice shall be given to the students.